Real-time Target Speaker Extraction

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Declaration

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Grant Booysen	**November 2026**
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Abstract

Abstract

Acknowledgements

Contents

Nomenclature

Abbreviations

BSRNN	Band-split recurrent neural network
LSTM	Long short-term memory
STFT	Short-time Fourier transform
TF Map	Time-Frequency Map
TSE	Target speaker extraction

Symbols

Chapter 1

Introduction

Chapter 2

Literature Review

2.1. Band-split Recurrent Neural Network (BSRNN)

Luo and Yu [1] propose the BSRNN model as a frequency-domain separation model that is designed for high sample rate audio, in which the input audio is transformed from its waveform representation into a time-frequency representation using the Short-Time Fourier Transform (STFT). The model then divides the input spectrum into multiple frequency bands that are modelled independently. The original use case for the BSRNN model is music source separation, but in this work, the model is adapted to allow for the extraction of a target speaker from a mixture of speech signals. The development from music source separation to the method of the target speaker extraction (TSE) approach that this work follows came from the following lineage: *(i)* music source separation was adapted to personalised speech enhancement (PSE) by Yu et al. [2]. The authors proposed target speaker extraction in the PSE task by appending a target speaker enrollment module that was designed to suppress any interference. PSE can be summarised as enrollment-conditioned extraction. *(ii)* Zhang et al. followed this framework and proposed a new module for the enrollment conditioning module, namely the Time-Frequency Map (TF Map) [3].

Chapter 3

Methodology

3.1. Baseline model architecture

3.1.1. Short-time Fourier transform

When speaking, the audio contains energy. The energy is not evenly distributed across frequencies, and in normal speech it is not evenly distributed across time either. The energy is concentrated in particular frequency bands by the unique resonances of the speaker's vocal tract, and is also concentrated in time by syllables, words and phrases. Audio models therefore want to leverage that structure, and require the input to be presented in a representation that allows for those concentrations to be observed. The waveform itself cannot represent that structure as it is a one-dimensional signal and is not aligned with the human perception of sound.

The short-time Fourier transform (STFT) produces that representation. The waveform is cut into short overlapping segments. These segments are used to compute the Fourier transform, which is used to decompose the signal into its frequency components. These overlapping segments are called frames, and the idea is to pass each of these segments through the Fourier transform by first multiplying the segment by a window function. The window function, which is typically a Hann window, tapers at the ends such that when multiplied, silences the ends of the segment to isolate the middle. After this a discrete Fourier transform is taken of each segment. The result is one complex number per frequency bin per segment: for a real input x, the coefficient at frequency bin f and frame t is

$$X(f,t) = \sum_{n=0}^{N-1} x[tH+n]\, w[n]\, e^{-j2\pi fn/N}, \tag{3.1}$$

where j is the imaginary unit ($j = \sqrt{-1}$), N is the window length in samples, H is the hop by which successive windows advance, and $w[n]$ is the window function. The hop is what makes the segments overlap: consecutive frames share $N - H$ samples, so a feature that falls near the edge of one frame sits comfortably inside the next. For the implementation in this work, a window size of $N = 512$ is selected, with a hop of $H = 128$ samples at $16\,\mathrm{kHz}$ gives a $32\,\mathrm{ms}$ window shifting by $8\,\mathrm{ms}$. These are the values Yu et al. [2] specify for their $16\,\mathrm{kHz}$

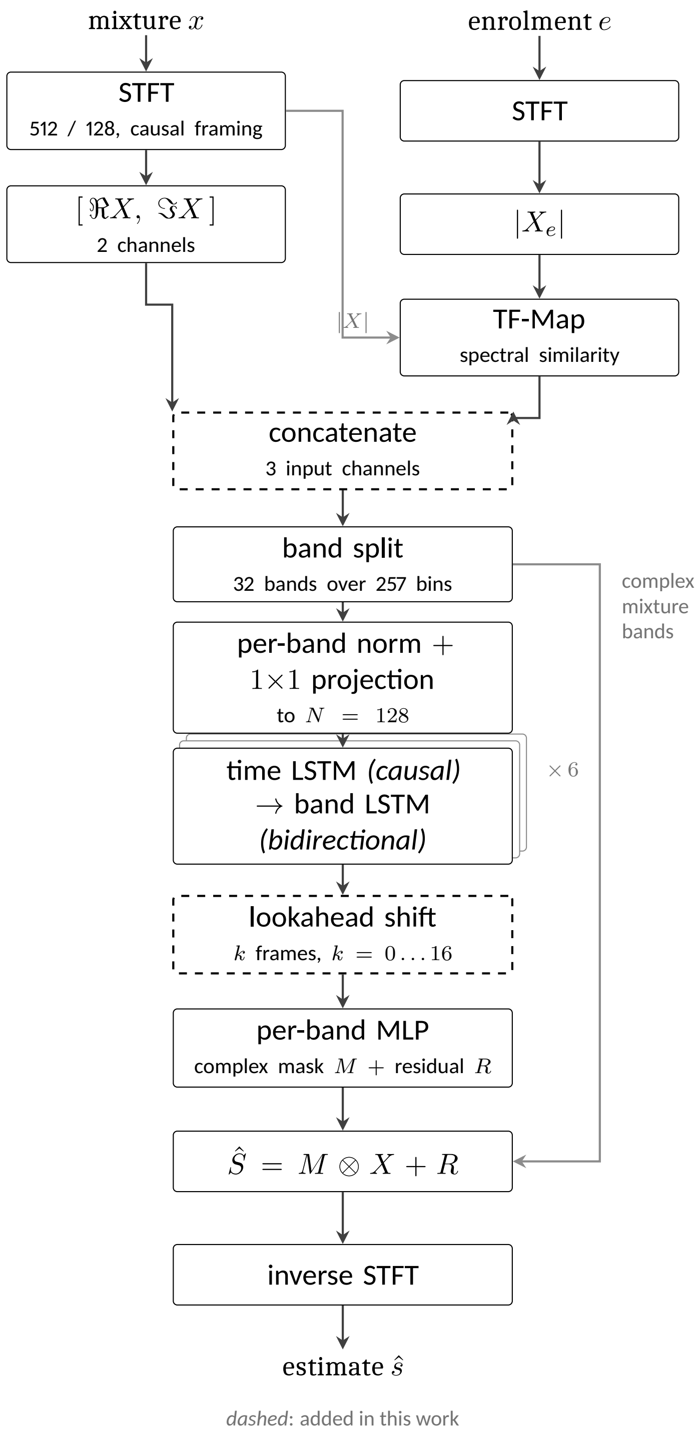

Figure 3.1: The extractor as implemented. Band splitting and the interleaved band-level and sequence-level modelling follow [1]; the mask-plus-residual estimator follows [2]; TF-Map conditioning follows [3]. The causal time path, the lookahead shift and the third input channel are this work's.

model. Secondly given the constraint of the streaming budget of $200\,\mathrm{ms}$ to $300\,\mathrm{ms}$, the framing consumes well under a quarter of that budget, and the remaining headroom allows for budget to be spent on the lookahead of Section 3.1.5. Finally, the approach in this work forms the frames without centring creating the causal framing. The reason for this is that centring each frame on its own timestamp makes frame t consume $N/2 = 256$ samples – $16\,\mathrm{ms}$ – of audio recorded *after* t. Since the objective is to create an online, streaming system and thus the model should not depend on future input. To account for the beginning samples of the audio, the signal is padded by $N - H$ samples at the start, which is a kind of cold start of a live system whose input buffer begins empty.

Each coefficient carries two quantities. Its magnitude $|X(f,t)|$ is how much energy the signal has in that frequency band at that time, and its phase $\angle X(f,t)$ is where in repetition of the waveform the signal is at that time. The magnitude is what the model uses to determine which parts of the mixture belong to the target speaker, and the phase is what the model uses to reconstruct the waveform from the modified magnitude. The complex spectrogram is kept separately, since the mask of Section 3.1.6 is applied to it rather than to the network's internal features.

3.1.2. Band plan

The spectrogram of Section 3.1.1 has 257 outputted frequency bins, each covering an equal slice of $31.25\,\mathrm{Hz}$. Treating all 257 alike would waste the model's capacity, because speech does not spread itself evenly across them. Most of the energy, and nearly all of the structure that distinguishes one voice from another, sits at the lower frequencies: the pitch of a voice and the resonances that give a vowel its identity all fall below about $4\,\mathrm{kHz}$, and energy falls away steadily above that. The bins near the top carry far less information and thus would be wasteful to model with the same level of detail as the bins near the bottom.

A band plan is the decision of how to group those bins. Bins are gathered into contiguous groups, called bands, and each band is then handled separately by the network: it gets its own normalisation and its own projection, so a band is compared against itself over time rather than against the loud low-frequency bands next to it. Grouping few bins per band keeps fine detail. Secondly grouping many bins into one band summarises coarsely but costs far less. A band plan therefore decides where the model's resolution is spent.

The plan in this work is 32 bands over the 257 bins: fifteen bands of $100\,\mathrm{Hz}$ covering $0\,\mathrm{kHz}$ to $1.5\,\mathrm{kHz}$, ten of $200\,\mathrm{Hz}$ covering $1.5\,\mathrm{kHz}$ to $3.5\,\mathrm{kHz}$, five of $500\,\mathrm{Hz}$ covering $3.5\,\mathrm{kHz}$ to $6\,\mathrm{kHz}$, and the remaining $6\,\mathrm{kHz}$ to $8\,\mathrm{kHz}$ as one band. Figure 3.2 shows it against a real mixture in the training set. The effect is a factor of twenty between the finest and coarsest resolution. This plan follows the implementation from Zhang et al. [3] for their $16\,\mathrm{kHz}$ target speaker extraction model and used by the challenge baseline built on it, but no reason is given there for these particular boundaries, and none is given in the papers it descends from – the original band splitting was designed for music [1], where the sources are instruments rather than voices. It is adopted here as the baseline so that this work's results are comparable with that lineage, and it is recorded as an unjustified inheritance rather than a motivated choice.

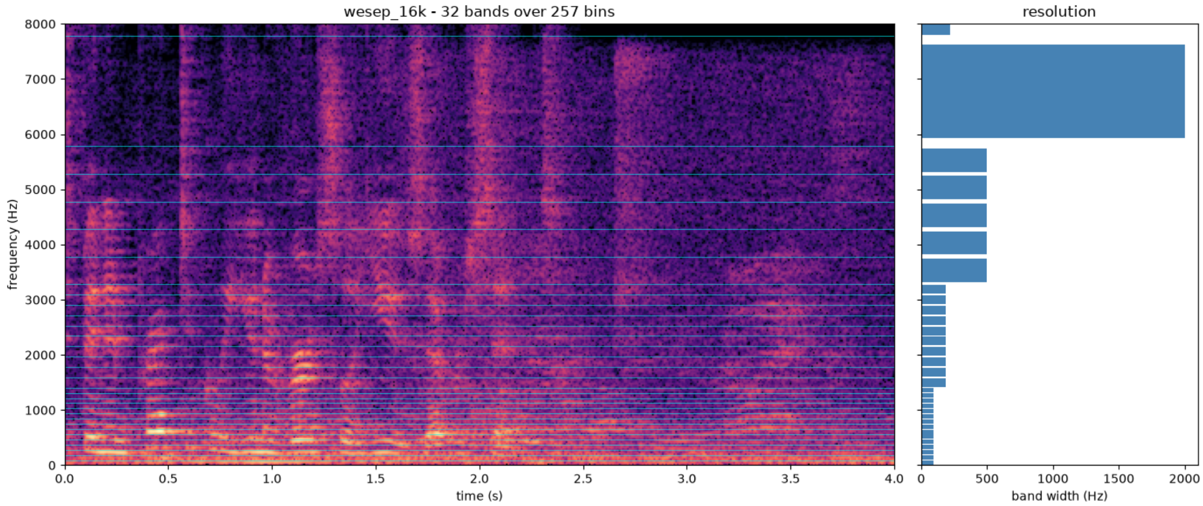

Figure 3.2: The band plan used here, shown against a four-second mixture from the training set. Left: the band boundaries drawn over the spectrogram, closely spaced at the bottom where the speech structure is and widening upwards. Right: the width of each band in hertz. The lowest fifteen bands are $100\,\mathrm{Hz}$ wide; the single highest spans $2\,\mathrm{kHz}$. This band plan follows the one used by Zhang et al. [3] for their $16\,\mathrm{kHz}$ model.

[GRANT: TODO - Possible experiment] Because the plan is a free parameter that nobody has justified, it is also implemented as a configurable table rather than fixed in code, with alternative plans available for comparison.

Why the spectrum stops at $8\,\mathrm{kHz}$. The audio in this work is sampled at $16\,\mathrm{kHz}$, meaning 16,000 amplitude measurements per second. When measuring a wave you need at least two measurements per cycle, one for the peak and one for the trough of the wave. Therefore the fastest possible wave that $16\,\mathrm{kHz}$ sampling can represent will complete 8,000 cycles per second. Anything faster cannot be distinguished from a slower wave passing through the same measured points. Therefore, the largest possible range that can be represented by the 257 bins have to span $0\,\mathrm{kHz}$ to $8\,\mathrm{kHz}$ rather than $0\,\mathrm{kHz}$ to $16\,\mathrm{kHz}$.

3.1.3. Per-band normalisation

The bands defined by Section 3.1.2 are not comparable with one another. Speech energy falls away by roughly $6\,\mathrm{dB}$ per octave, so the lowest bands carry orders of magnitude more energy than the highest. Therefore, if all 257 bins were scaled together, there would be a severe imbalance where the high bands would be underrepresented. This would then hurt the performance of the model as the high bands would be diluted and would contribute almost nothing to the gradient, and the model would in effect only learn to separate the bottom of the spectrum. Giving each band its own scaling is what prevents that. A band at $6\,\mathrm{kHz}$ is then judged against its own typical level rather than against the bands beneath it.

This module does two things to each band independently, following [2]. First it normalises the band, using a layer normalisation applied across the channels of that band. Secondly, because the bands must be the same size before they can be stacked together, a projection is required to map the band to a fixed width. Since there are varying sized and independent bands, each band has its own projection, which is a learnable linear layer that maps the band to a fixed width of $N = 128$ values. Every band is given the same structure but its own weights, so nothing is shared between bands and each is free to learn a representation suited to its own frequency range. The result is a stack of 32 bands, each with 128 features, which is then passed to the next module.

Normalisation is applied across the values of a band *within a single frame*. Each frame is therefore normalised using only itself. This is a deliberate design choice that opposes Yu et al. [2], who use batch normalisation for their online configuration. The opposition is for the following reasons: *(i)* although batch normalisation will work for online models as indicated by Yu et al. and is causal, there are two different operations of batch normalisation: training and inference. During training, the mean and variance are computed from the current batch, but during inference, these statistics are computed from the entire training set. The function that is optimised during training is therefore not the function that is deployed. *(ii)* That gap is unusually wide here. A batch is twelve four-second crops drawn at random from a corpus spanning a range of reverberation times, signal-to-interference ratios and noise levels, and in which roughly three crops in ten contain no target speech at all. The statistics of a batch therefore depend on what was drawn, and a silent crop normalised alongside loud ones is not normalised as it would be at inference. Because normalisation is per band, the effect is worst in the high bands, where the energy is smallest and the variation between batches is largest relative to the signal. *(iii)* Single frame normalisation carries no state and is unaffected by the amount of audio processed, so a four-second training chunk is normalised exactly as a stream that has been running for a minute.

3.1.4. Residual LSTM block

The idea of the residual LSTM block is to model the sequential structure of speech such that the model can leverage patterns that unfold over time. There are two kinds of sequential structures that are modelled here. The first is the sequence over time, which is the natural order of speech. The second is the sequence over frequency. Think of the time being unidirectional, as the speech only makes sense to unfold forward in time, but the frequency is bidirectional, as all of the bands of a frame are available at the same instant and can be traversed in either direction. The block is used for both kinds of modelling, with the same structure but different parameters.

The recurrent layer is what carries information forward where this work leverages a long short-term memory layer, or LSTM. This is a recurrent layer with an internal mechanism for deciding what to keep in that summary and what to discard, which is what allows it to retain information over hundreds of frames rather than only the last few. This structure also suits the deployment target directly. At inference time, since the model is running in a streaming fashion, the LSTM's internal state is preserved between frames and information can be carried forward indefinitely. The LSTM is therefore a natural fit for the task, and it is used in the same way for both the time and frequency sequences.

The block is made of three steps applied in order, following [1]. The input is first normalised, then passed through the LSTM, and the LSTM's output is then mapped back to the input's width by a learned projection. The final addition is a residual connection [4], and is inspired by the original ResNet [4] architecture. This allows the gradient an unobstructed path back through the computation stack of all the blocks, which matters because six of these blocks are stacked in Section 3.1.5. This is important as the LSTM is a deep recurrent layer, and the gradient can vanish or explode as it passes through the LSTM's internal state. The residual connection allows the gradient to bypass the LSTM entirely, which is what allows the model to train when the gradient vanishes within the LSTM.

[GRANT: TODO] - Still might need to increase the size of the model so this is not final!!!

What is used here. The LSTM has a single layer with a width of 192, taken from Yu et al. [2], whose $16\,\mathrm{kHz}$ model specifies that width. This is deliberately narrower than the reference implementation of the same architecture, which uses twice the feature width, or 256. Because the LSTM's width differs from the 128 features entering the block, the projection is not optional bookkeeping: it is what makes the addition dimensionally possible at all.

The block is written to be indifferent to what its sequence axis represents. It receives a batch of sequences and processes them; whether those sequences run along time or along frequency is decided by the caller. This is what allows the same block to be used for both kinds of modelling in Section 3.1.5, with one critical difference between the two uses. Along time the LSTM runs forward only, because a backward pass over time reads frames that have not yet arrived. Along frequency it runs in both directions, which costs nothing in latency, since all the bands of a frame are available at the same instant.

3.1.5. Band and sequence model

3.1.6. Estimation module

3.1.7. Time-frequency map

3.1.8. Objective function

The objective has to express two different requirements, because the training data contains two different situations. Where the target speaks, the output should match the target signal. Where the target is silent as the result of either the trial contains no target speech at all, or because the four-second crop happened to land in a gap between the target's utterances, there is no signal to match, and the correct output is silence. A single loss cannot serve both use cases and so the objective is therefore split, and each crop is scored by the branch that applies to it.

Throughout, x is the mixture, s the target reference and $\hat{s}$ the model's output, all in the time domain over one training crop.

Target present The first term measures how much of the output is the target and how much is everything else, following [5]. Importantly, the loss needs to be invariable to the overall volume as if the $\hat{s}$ is of the correct shape, it should not be penalised for being loud or quiet. Therefore the first step is to find the single gain that best explains the output as a scaled copy of the target, which will be calculated as a projection,

$$\alpha = \frac{\langle \hat{s},\, s \rangle}{\|s\|^2}, \qquad s_{\text{proj}} = \alpha s, \tag{3.2}$$

and then comparing the energy of that scaled target against the energy of whatever the output contains beyond it. The loss is modelled after the Scale-Invariant Signal-to-Distortion Ratio (SI-SDR). The loss is therefore,

$$L_{\text{pres}} = -10 \log_{10} \frac{\|s_{\text{proj}}\|^2}{\|\hat{s} - s_{\text{proj}}\|^2 + \tau \|s_{\text{proj}}\|^2}. \tag{3.3}$$

Lower is better. The loss is secondly in terms of decibels (dB), which is where the logarithm comes from. The τ term in the denominator is a floor. Without it a perfect output would score $-\infty$, and the gradient would keep rewarding refinement of crops that are already essentially solved. With $\tau = 10^{-3}$ the best attainable score is $-30\,\mathrm{dB}$. This constant follows the suggestion from CARTSE [5]. The reasoning is that an output whose residual error already sits $30\,\mathrm{dB}$ below the target is close enough to be treated as solved, so it would be wasteful to keep spending gradient on it. Two properties of this measure should be stated plainly, because they are what the second term exists to compensate for. The first is that everything the output contains beyond the target is collected into one denominator. Residual interfering speech, residual background noise and artefacts the model itself introduced all count identically per unit of energy, although they are not equally damaging: a fragment of the interferer left in the output can be transcribed downstream as words the target never said, whereas a diffuse artefact of the same energy usually cannot. The measure has no way to express that difference. The second is that it is an energy ratio summed over the whole crop, so it is dominated by whatever is loudest in it. Speech energy sits overwhelmingly in the low frequencies and in vowels, which means a model can score well on (3.3) while reconstructing fricatives and the upper spectrum poorly. Those are the parts that carry consonant identity, and therefore much of what makes speech intelligible rather than merely present.

Multi-resolution spectral term. Equation (3.3) is a single number computed over the whole waveform, which makes it ignore the location of errors in the spectrum. Since it cannot express *where* in the spectrum the error lies, a model that reconstructs the loud low bands well and the quiet high bands poorly is barely penalised and secondly since being scale-invariant, it does not constrain the output's absolute loudness at all. The idea is to use a loss that considers multiple resolutions of s and $\hat{s}$. By considering different granularity of the spectrogram, the model is encouraged to reconstruct both the low and high frequency bands well. A spectral term allows for this following [2]:

$$L_{\mathrm{MR}} = \frac{1}{I}\sum_{i=1}^{I}\left(\left\| |S_i|^p - |\hat{S}_i|^p \right\|_1 + \left\|\Re S_i - \Re \hat{S}_i\right\|_1 + \left\|\Im S_i - \Im \hat{S}_i\right\|_1\right), \tag{3.4}$$

where S_i and $\hat{S}_i$ are the spectrograms of the target and the output under the ith of I analysis window lengths, and each norm is averaged over its time-frequency coefficients.

Three choices inside (3.4) are worth stating. The exponent $p = 0.3$ compresses magnitudes, which will highlight the quieter components relative to loud ones so that an error in a low-energy region is not diluted just because it is quite. Without the compression the term would be dominated by the same loud low bands that already dominate (3.3). The real and imaginary terms are left uncompressed and they are what make the term sensitive to phase: a magnitude-only loss would be indifferent to the phase that the complex mask of Section 3.1.6 is free to change. Finally the term is evaluated at several window lengths rather than one, because a single analysis window can hide errors at its own scale.

The window lengths used are 8, 16, 32 and $64\,\mathrm{ms}$. This is with respect to the model's own $32\,\mathrm{ms}$ framing. The model builds its output by masking $32\,\mathrm{ms}$ frames and overlapping them, so the places that most errors would occur can be expected to exist at frame boundaries. The $8\,\mathrm{ms}$ and $16\,\mathrm{ms}$ windows resolve this and the $64\,\mathrm{ms}$ window catches harmonic structure that the short ones blur.

Target absent When there is no target speaker present in the mixture audio, the model should output silence and suppress all other noise. The model should therefore penalise any output that is not silent. This is modelled after a modified version of the absolute level loss from CARTSE [5], where in this works the loss now contains the denominator term.

$$L_{\text{abs}} = 10 \log_{10} \frac{\|\hat{s}\|^2 + \tau \|x\|^2}{\|x\|^2}. \tag{3.5}$$

Including the denominator term makes the loss relative to the input, which is directly readable: $0\,\mathrm{dB}$ means the output is as loud as the mixture, that is, the model did nothing; $-10\,\mathrm{dB}$ means the interfering speech was suppressed by $10\,\mathrm{dB}$; and the same τ floors the term at $-30\,\mathrm{dB}$, beyond which the output counts as silent. A positive value means the model amplified the mixture, which is a direct failure. The inclusion of the denominator also allows for scale invariance, which puts the loss in the same units as the loss for the target present case.

Combining the two. Each term is averaged over the crops it applies to, and the two averages are weighted against one another:

$$L = (1 - w) \underset{\text{target present}}{\mathrm{mean}} \left[L_{\text{pres}} + w_{\mathrm{m}} L_{\text{MR}} \right] + w \underset{\text{target absent}}{\mathrm{mean}} \left[L_{\text{abs}} \right]. \tag{3.6}$$

Averaging within each branch rather than over the batch as a whole matters more than it appears to. Under a single batch-wide mean, the share of the gradient belonging to the silent crops would be whatever fraction of the batch happened to be silent, so the balance between the two requirements would fluctuate from step to step on the luck of the draw. Under (3.6) that balance is fixed by w.

Which branch a crop belongs to is decided from the audio of the cropped target itself, not from the trial's label. A trial in which the target does speak can still yield a crop containing no target speech, and this occurs in $5.8\,\%$ of crops. Routing such a crop to the present branch would put an all-zero reference into (3.2), where α becomes $0/0$; the implementation asserts against this rather than relying on the labels agreeing.

Table 3.1: Departures from the objective of [5] as published. Each is recorded with its reason in the project's decision log.

	As published	As used here
1. Floor in (3.3)	On $\tau\|s\|^2$, which is not scale-invariant	On $\tau\|s_{\mathrm{proj}}\|^2$, so that gain cancels
2. Scale of (3.5)	Unnormalised, so the value shifts with input loudness	Divided by $\|x\|^2$, so $0\,\mathrm{dB}$ means "did nothing"
3. Reduction	One mean over the whole batch	A mean within each branch, so the balance is set by w alone
4. Number of weights	A separate weight on the silent term	Folded into w; under branch-wise means the two are one degree of freedom
5. Silence weight	Twice the absent rate of the source corpus	$w = 0.458$, the same emphasis at this project's measured 0.297
6. Windows in (3.4)	All at or above the model's frame length	8, 16, 32 and $64\,\mathrm{ms}$, straddling it

Chapter 4

Experiments

Chapter 5

Results

Chapter 6

Conclusion

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Appendix A

[Appendix title]